

PROCESS INTEGRATION TRAINING

FLOWCHART

VERSION 3.7.4

Valid from March 2026

BATCH INFORMATION			
NAME OF OWNER :	EKL	MASK SET :	PDM 16
NAME OF MENTOR :	Mario	MASK BOX :	269A and 269 B
RUN NUMBER :	PIT group _____	DIE SIZE :	10 by 10 mm
WAFER AMOUNT :	6	START DATE :	/ /
SUBJECT TO PCC :	no	PCC APPROVED :	n.a.

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GENERAL RULES

CLEANROOM BEHAVIOUR

- ❶ Always follow the "**Security and Behavior**" rules when working in the EKL laboratories.
- ❷ Always handle wafers with care during processing. Use cleanroom gloves and work as clean as possible!
- ❸ Use cleanroom gloves when working with vacuum equipment. Do not touch the inside or carriers with bare hands.
- ❹ Always check equipment and process conditions before starting a process. Do **NOT** make unauthorized changes!
- ❺ Directly notify the responsible staff member(s) when there are problems with the equipment (like malfunction or contamination). Flip the status card on the machine over to **DOWN** to warn other users. Also change the status of the system to **DOWN** in the "[NanoLabNL Information System \(NIS\)](#)" system.
- ❻ **DO NOT TRY TO REPAIR OR CLEAN EQUIPMENT YOURSELF**, and **NEVER** try to refresh a contaminated etch or cleaning bath! Only authorized staff members are allowed to do this.

WORKING WITH CMOS INCOMPATIBLE MATERIALS

- ❶ Substrates, layers and chemicals which are not CMOS compatible may cause contamination of bathes, equipment, wafer boxes, etc.. Using these materials in the class 100 and SAL cleanroom without permission is **FORBIDDEN**.
- ❷ The use of CMOS incompatible materials for processing in the class100 and SAL cleanroom must **ALWAYS BE EVALUATED** and **APPROVED** by your mentor and the EKL contamination officer.
- ❸ Wafers that are contaminated may **NEVER** be processed in any of the bathes or equipment without permission. Special precautions may be required, like the use of a separate container, a special substrate holder or a wafer carrier.
- ❹ You **MUST** work according to the rules described in the **Preventive Cross Contamination (PCC)** document, available on the "[EKL Sharepoint webpage](#)", and the **Material Compatibility** database from the "[NanoLabNL Information System \(NIS\)](#)" system.

CLEANING OF WAFERS

Wafers must always be cleaned before performing a **COATING, FURNACE, EPITAXY** or **DEPOSITION** step if they were stored for 4 hours or more.

Use the correct cleaning bathes:

- | | |
|---|---|
| • Acetone | ⇒ To remove photoresist that is not used as a mask for ion implantation or plasma etching. |
| • Tepla stripper | ⇒ To remove ion bombarded photoresist after an ion implantation or plasma etching step. |
| • HCl (10%) - Si (cleaning*) | ⇒ For importing wafers that were processed outside the EKL class100 cleanroom, and that were never in contact with metal layers. APPROVAL for importing these is needed. |
| • HNO ₃ 99% (Si) | ⇒ To remove organic material from wafers which were never in contact with metal layers. |
| • HNO ₃ 69.5% (Si) | ⇒ To remove (possible) metal particles caused by wafer handling (by equipment). It is never allowed to use this bath for wet etching or stripping of metal layers! |
| • HNO ₃ (99%)(green metals*) | ⇒ For importing wafers that were processed outside the EKL class100 cleanroom, and that at some point were in contact with "green" metals. APPROVAL for importing is needed. |
| • HNO ₃ 99% (green metals) | ⇒ To remove organic material from wafers which are or have been in contact with "green" metals (e.g.: Al, Al (1%Si), Ti, TiN, Zr, ZrN, Nb, etc.). Do not use for removing organic materials from wafers with Mo, as this metal will be etched by the acid. |

Note: The above described cleaning procedures are only valid for CMOS compatible wafers with CMOS compatible materials on them. **For all other wafers follow the PCC rules and check the Material Compatibility database.**

Wafers do **NOT** have to be cleaned **after** a furnace, epitaxy or deposition step if the next process step will be performed within 4 hours, unless the wafers are covered with particles.

FURNACE RESTRICTIONS

Wafers that are covered with an organic material (e.g. photoresist) may **NEVER** be processed in **any** of the furnaces. Wafers from which a metal layer was removed by etching are only allowed in tube A3 (**green** metals) or A4 (**red** metals). Wafers covered with a metal layer are only allowed in tube A3 or A4 for alloying at a **temperature below the melting point**.

MEASUREMENTS

Always perform all the measurement and inspection steps, and **write down the results in your journal and in the logbooks that can be found at some of the equipment**. The results are used to monitor the processes and/or equipment.

It is possible to measure directly on your (CMOS compatible) process wafers with the following Class 100 equipment:

- The WOOLLAM ellipsometer, the FR-scanner/Reflectometer and the KEYENCE VK-X250 microscope. The first 2 systems are used for thickness measurements of transparent layers, and the third system is used for 3D surface metrology. The measurements are non-destructive and without contact to the wafer surface.
- The Dektak 8 surface profilometer. This system is used for step height measurements. In this case a needle will physically scan over the wafer surface, which can be destructive for structures. It is a contact measurement.
- The Hitachi Regulus 8230 SEM, which can be used for inspection and for width, depth or thickness measurements.

Note:

- After certain measurements **cleaning of the measured wafer** may be required for further processing.
- An extra wafer must be processed when other measurements are required (like sheet resistance and junction depth measurements). These wafers can not be used for further processing.

STARTING MATERIAL

Use **SINGLE SIDE** polished **LOW RESISTIVITY (L_{RES})** wafers,
with the following specifications:

Type: n / phosphorus

Orientation: <100>

Resistivity: 1-5 Ωcm

Thickness: $525 \pm 15 \mu\text{m}$

Diameter: 100 mm

- Wafers that are acquired **new and unused** out of a box from the Engineer On Duty (or another staff member) do **NOT** need to be cleaned before a coating, furnace or deposition step. These wafers were cleaned by the supplier and can not be cleaned better than that at EKL.
- For an epitaxy step it will be necessary to remove the native oxide in the **Marangoni wet etch & drying system for Si wafers**, as a bare Si surface is required for this process.

1. COATING

Front side

Use the coater station of the EVG120 system to coat the wafers with photoresist. The process consists of:

- A treatment with HMDS (hexamethyldisilazane) vapor, with nitrogen as a carrier gas, to improve resist adhesion.
- Spin coating of Shipley SPR3012 positive resist, dispensed by a pump.
- A soft bake (a.k.a. pre-bake) at 95 °C for 90 seconds.
- An automatic Edge Bead Removal (EBR) with a solvent.

Always check the relative humidity ($48 \pm 2\%$) in the room before coating, and follow the instructions for this equipment.

Default use program "**1 - Co - 3012 - zero layer**", which has a larger Edge Bead Removal (EBR) than standard, or use program "**1 - Co - 3012 - 1.4 - noEBR**" when the EBR option is out of order.

- Wafers that are coated with **unexposed** resist must always be stored on the rack in the lithography room. Do **NOT** store them in your cupboard, to prevent unintended exposure by white light (which contains UV-light).
- After coating of your wafers with photoresist, they can be easily stored up to 4 weeks. They will then still be OK to be exposed with your mask. There is no need to remove the resist, clean the wafers and re-coat them.

2. ALIGNMENT AND EXPOSURE

Front side

Processing will be performed on the ASML PAS5500/80 automatic wafer stepper.

Follow the operating instructions from the manual when using this machine.

Expose **mask COMURK** with job **Litho\10-10 comurk0.0**. Use the correct exposure energy (check the energy table). This results in stepper alignment markers and verniers for wafers which will not get an Epitaxy (EPI) layer.

- After exposure the photoresist must preferably be developed as soon as possible, but **within 2 hours!**

3. DEVELOPING

Front side

Use the developer station of the EVG120 system to develop the wafers. The process consists of:

- A post-exposure bake at 115 °C for 90 seconds, to prevent **standing waves** on the sidewalls of the resist structures.
- Developing with Shipley MF322 with a single puddle process.
- A hard bake (a.k.a. post-bake) at 100 °C for 90 seconds.

Always follow the instructions for this equipment.

Use program "**1 - Dev - SP**".

- After development the wafers can be stored in your cupboard as usual. Exposure to white light does not matter.

4. INSPECTION

Front side

Visually inspect the wafers through a microscope:

- Check if the correct mask is exposed.
- Check if there are no resist residues on the exposed areas; these are NOT allowed.

5. WAFER NUMBERING

Front side

Use the glass pen in the lithography room to mark the wafers with the **BATCH** and **WAFER** number.

Write the numbers in the photoresist, just above the waferflat. Always do this after exposure and development !

It is **NOT** allowed to use a metal pen or a scribe (pen with a diamond tip) for this purpose.

6. PLASMA ETCHING: Alignment markers (URK's) into Silicon

Front side

Use the Trikon Omega 201 plasma etcher.

Follow the operating instructions from the manual when using this machine.

It is **not** allowed to change the process conditions and times from the etch recipe!

Use sequence: **URK_NPD** (with a platen temperature of **20 °C**) to etch 120 nm deep stepper alignment markers into Si.
The sequence consists of 2 etch recipes: URK_BRK (to etch native SiO₂) and URK_BULK (to etch Si).

Process conditions from chamber recipe URK_BRK:						
Step	Gasses & flows	Pressure	Platen RF	ICP RF	Platen temp.	Etch time
1. Breakthrough	CF ₄ /O ₂ = 40/20 sccm	5 mTorr	60 W	500 W	20 °C	0'15"

Process conditions from chamber recipe URK_BULK:						
Step	Gasses & flows	Pressure	Platen RF	ICP RF	Platen temp.	Etch time
1. Bulk etch	Cl ₂ /HBr = 80/40 sccm	60 mTorr	20 W	500 W	20 °C	0'40"

PIT comment:

After most process steps a visual and/or measurement inspection must be performed before continuing with the next step, as faults usually can not be repaired at a later stage. After alignment marker etching there is no inspection however, as we can not see if the etch result is OK. Silicon is etched and in silicon is landed. We see no difference.

7. LAYER STRIPPING: Photoresist

Front side

Strip resist Use the Tepla Plasma Gigabatch 360 system to remove the photoresist in an oxygen plasma.
Follow the instructions specified for the Tepla stripper, and use the quartz carrier.
Use **program 1**: 1000 watts power and automatic endpoint detection + 2 min. overetching.

8. CLEANING: HNO₃ 99% and 69.5%

Both sides

Clean 10 minutes in fuming nitric acid at ambient temperature. This will dissolve organic materials.
Use wet bench "HNO₃ 99% (Si)" and the carrier with the white dot.

Rinse Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.

Clean 10 minutes in concentrated nitric acid at 110 °C. This will dissolve metal particles.
Use wet bench "HNO₃ 69,5% 110C (Si)" and the carrier with the white dot.

Rinse Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.

Dry Use the "Avenger Ultra-Pure 6" rinser/dryer with the standard program, and the white carrier
with a red dot.

PIT explanation:

HNO₃ 99% is used to remove organic contaminants from wafers. This material usually consists of residues that are left behind after photoresist removal, but can also be skin particles or hairs (when wafers are not handled correctly). Other materials (including metals) will not be affected or get damaged. During cleaning the bath must be closed as HNO₃ 99% is hygroscopic and tends to absorb moisture from the air, thus diluting itself. Also dilution takes place by using a wet carrier and handle. When HNO₃ 99% gets diluted, it starts dissolving metals.

HNO₃ 69.5% (at 110°C) is used to remove metal contaminants. At this stage of processing there should not be any metal on your wafers, but wafers can pick up metal particles when they are handled in other equipment (from metal robot hands, chucks, etc.). These need to be removed before processing the wafers in furnaces or other high temperature equipment, to prevent contamination of the equipment and diffusion of metal into the silicon.

A minimum time of 10 minutes is required to clean the wafers. Leaving wafers longer in the bath does not cause damage to the wafer or the layers on the wafer.

9. WET OXIDATION: 300 nm thermal SiO₂ on wafers 1 and 2

Both sides

Furnace tube: **A2** Program name: **EKL_WET_1000C** Total time: ~ 190 min + oxidation time

PROCESS	TEMPERATURE (in °C)	GASSES & FLOWS (in liter/min)	TIME (in minutes)	REMARKS
boat in	700	nitrogen: 10.0	5	
heat up	+10 °C/min	nitrogen: 5.0	~ 50	
stabilize	1000	nitrogen: 5.0	5	
start torch	1000	oxygen: 4.0 hydrogen: 6.0	1.5	
oxidation	1000	oxygen: 4.0 hydrogen: 6.0	to be calculated	variable command
stop torch	1000	oxygen: 4.0 hydrogen: 0.0	1	
purge	1000	nitrogen: 6.0	0.5	
post anneal	1000	nitrogen: 5.0	2	
cool down	-10 °C/min	nitrogen: 5.0	~ 120	wait for operator
boat out	700	nitrogen: 6.0	5	

Oxidation time:

Use an oxide growth calculator to calculate the oxidation time that is needed to reach the desired thickness. Check out cleanroom.byu.edu/OxideTimeCalc or www.lelandstanfordjunior.com/thermaloxide.html. Then input the calculated time at the variable command line of the **EKL_WET_1000C** recipe.

10. LPCVD DEPOSITION: 300 nm TEOS on wafers 3 and 4

Both sides

Furnace tube: **C1** Program name: **TEOS_std**

Follow the instructions for the LPCVD furnace when using this equipment.

Process conditions from recipe TEOS_300nm:			
Gasses & flows	Pressure	Temperature	Time
TEOS = 80 ml/min	200 mTorr	715 °C	variable command

- Note:**
- The layer thickness depends on the used deposition time, which can be calculated with the deposition rate that was measured during recent recipe usage. This can be found in the **excel log file** for the system.
 - An extra test wafer can be deposited for measurements and etch tests.

Alternative PIT step if LPCVD tube C1 is unavailable:

Use program "**SiN-M**" in LPCVD furnace **F3** to deposit a Silicon Nitride (SiN) film instead of a TEOS layer. Use the Excel log file on the PC in front of the Epsilon reactor to calculate the time to deposit a 300 nm thick film. **Remember to use the boat for clean wafers and that later a SiN etch recipe will be needed to pattern this layer.**

11. PECVD DEPOSITION: 300 nm Silicon Oxide on wafers 5 and 6

Front side

Use the Novellus Concept One PECVD reactor.

Follow the operating instructions from the manual when using this machine.

It is **not** allowed to change the process conditions and time from the deposition recipe!

Use recipe "**.xxx_siostd**" to deposit a 300 nm thick SiO₂ layer.

Process conditions from recipe .xxx_siostd:					
Gasses & flows	Pressure	HF power	LF power	Temperature	Time
SiH ₄ /N ₂ O/N ₂ = 210/6000/3150 sccm	2.2 Torr	1000 W	0 W	400 °C	variable

- Note:**
- The layer thickness depends on the station deposition time (SDT), which can be calculated with the deposition rate that was measured during recent recipe usage. This can be found in the logbook of the system.
 - An extra test wafer can be deposited for measurements and etch tests.

Alternative PIT step:

When the Novellus temperature is set at 350°C (instead of 400°C), then a 300 nm pecvd **TEOS** oxide layer will be deposited during the training to prevent the extra 30 min. heating up time. Use recipe **".xxxnmTEOS"**.

Process conditions from recipe .xxxnmTEOS:					
Gasses & flows	Pressure	HF power	LF power	Temperature	Time
TEOS = 1.800 ^{ml} / _{min} / O ₂ = 5.00 slm	2.2 Torr	500 W	500 W	350 °C	variable

12. MEASUREMENT: Oxide thickness (and Nitride thickness)

Front side

Use the Woollam Ellipsometer system for layer thickness measurements on 1 wafer from each process.
Follow the operating instructions from the manual when using this equipment.

For Silicon Oxide measurements use:
SiO₂ on Si (Thermox) (50nm-10µm) 5 pts

For LPCVD Silicon Nitride measurements use:

Expected layer thickness:	Thermal SiO ₂	⇒ 300 ± 10 nm	
	LPCVD TEOS	⇒ 300 ± 10 nm	(or LPCVD SiN ⇒ 300 ± 10 nm)
	PECVD SiO ₂	⇒ 300 ± 10 nm	(or PECVD TEOS ⇒ 300 ± 10 nm)

13. COATING

Front side

Use the coater station of the EVG120 system to coat the wafers with photoresist. The process consists of:

- A treatment with HMDS (hexamethyldisilazane) vapor, with nitrogen as a carrier gas, to improve resist adhesion.
- Spin coating of Shipley SPR3012 positive resist, dispensed by a pump.
- A soft bake (a.k.a. pre-bake) at 95 °C for 90 seconds.
- An automatic Edge Bead Removal (EBR) with a solvent.

Always check the relative humidity (48 ± 2 %) in the room before coating, and follow the instructions for this equipment.

Default use program **"1 - Co - 3012 - 1.4"**,
or use program **"1 - Co - 3012 - 1.4 - noEBR"** when the EBR option is out of order.

14. ALIGNMENT AND EXPOSURE

Front side

Processing will be performed on the ASML PAS5500/80 automatic wafer stepper.
Follow the operating instructions from the manual when using this machine.

Expose **mask CO** with job **Diesize_10mm\g10a-1 (layer ID 1)**. Use the correct exposure energy (check the energy table).

15. DEVELOPING

Front side

Use the developer station of the EVG120 system to develop the wafers. The process consists of:

- A post-exposure bake at 115 °C for 90 seconds, to prevent **standing waves** on the sidewalls of the resist structures.
- Developing with Shipley MF322 with a single puddle process.
- A hard bake (a.k.a. post-bake) at 100 °C for 90 seconds.

Always follow the instructions for this equipment.

Use program **"1 - Dev - SP"**.

16. INSPECTION

Front side

Visually inspect the wafers through a microscope:

- Check if the correct mask (from the correct mask set) is exposed.
- Check if there are no resist residues on the exposed areas; these are NOT allowed.
- Check the overlay of the exposed pattern (if the mask was aligned to a previous pattern on the wafer).
- Check the linewidth of the structures.

17. WET ETCHING: 300 nm oxide on **wafers 1, 3 and 5**

Both sides

Moisten	Rinse for 1 minute in wet bench "H ₂ O/Triton X-100 tbv BHF 1:7". Use the carrier with the blue dot. The bath contains 1 ml Triton X-100 per 5000 ml deionized water.
Etch	Use wet bench "BHF 1:7 (SiO ₂ -ets)" at ambient temperature, and the carrier with the blue dot. The bath contains a buffered HF solution.
Time	Etch until the windows on the front side are hydrophobic, plus an extra 30 seconds. The required etch time depends on the layer thickness and composition. The etch rate of thermally grown oxide is 1.3 ± 0.2 nm/s at 20 °C.
Rinse	Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.
Dry	Use the "Avenger Ultra-Pure 6" rinser/dryer with the standard program, and the white carrier with a black dot.
Inspection	Visually, through a microscope: All the windows must be open and the hydrophobic test may be applied.

18. PLASMA ETCHING: 300 nm Silicon Oxide on **wafers 2, 4 and 6**

Front side

Use the Drytek Triode 384T plasma etcher.

Follow the operating instructions from the manual when using this machine.

It is **not** allowed to change the process conditions from the etch recipe, except for the etch time!

Use recipe **STDOXIDE** to etch the silicon oxide layer. **Set the etch time to 33 seconds.**

Process conditions from recipe STDOXIDE:					
Step	Gasses & flows	Pressure	RF power	He pressure	Etch time
1. bulk etch (RIE)	C ₂ F ₆ /CHF ₃ = 36/144 sccm	180 mTorr	300 W	12 Torr	variable

INSPECTION: No Silicon Oxide residues are allowed on the etched areas.

Alternative PIT step:

If on wafer **3 and 4** LPCVD Silicon Nitride was deposited instead of TEOS, then for plasma etching of that layer recipe **STDSIN** must be used with an etch time of **55 seconds**.

Process conditions from recipe STDSIN:					
Step	Gasses & flows	Pressure	RF power	He pressure	Etch time
1. bulk etch (RIE)	C ₂ F ₆ = 65 sccm	130 mTorr	250 W	8 Torr	variable

INSPECTION: No Silicon Nitride residues are allowed on the etched areas.

19. PLASMA ETCHING: remove 300 nm TEOS oxide from the backside of **wafer 4** (only if LPCVD TEOS was deposited) **Backside**

Use the Drytek Triode 384T plasma etcher.

Follow the operating instructions from the manual when using this machine.

It is **not** allowed to change the process conditions from the etch recipe, except for the etch time!

Use recipe **STDOXIDE** to etch the oxide layer. **Set the etch time to 33 seconds.**

Process conditions from recipe STDOXIDE:					
Step	Gasses & flows	Pressure	RF power	He pressure	Etch time
1. bulk etch (RIE)	C ₂ F ₆ /CHF ₃ = 36/144 sccm	180 mTorr	300 W	12 Torr	variable

INSPECTION: No oxide residues are allowed on the etched areas.

PIT explanation:

A TEOS layer must be removed from the **backside** of the wafer before performing a metallization step. In that step the wafer will be heated up to 350 °C, which will cause outgassing of the TEOS layer. As a result the wafer will start to float on the electrode, which may cause it to fall off as it is only kept in place by gravity. This is common for both **lpcvd** as **pecvd** TEOS, and thus it is restricted to bring TEOS layers in direct contact with the heated electrode. This also applies for other machines that are equipped with a heated electrode (e.g. Novellus).

20. LAYER STRIPPING: Photoresist on wafers 1, 3 and 5 in Acetone

Front side

Strip resist Dissolve the photoresist in acetone at 40 °C. Time = 1 minute.
Use wet bench "Aceton 40°C" and the carrier with the two red dots.

PIT explanation:

The easiest (and quickest) way to remove a photoresist layer from a wafer is to submerge it in acetone (40°C), which will quickly dissolve the layer. This **does NOT work** for resist layers that have been **bombarded with ions** however, which happens **during ion implantation and during plasma etching**. The top layer of the resist carbonizes which causes that it becomes insoluble. When submerged in acetone the layer will start to wrinkle and sometimes partly come off, causing resist flakes to float around in the bath. This then needs to be refreshed. A wrinkled resist layer can still be removed from your wafer with an O₂ plasma.

21. LAYER STRIPPING: Photoresist on wafers 2, 4 and 6 in an oxygen plasma

Front side

Strip resist Use the Tepla Plasma Gigabatch 360 system to remove the photoresist in an oxygen plasma.
Follow the instructions specified for the Tepla stripper, and use the quartz carrier.
Use **program 1**: 1000 watts power and automatic endpoint detection + 2 min. overetching.

PIT explanation:

Resist layers that have been bombarded with ions **MUST** be removed with an oxygen plasma. This process is called **ashing** and takes more time. Ashing can also be used for removing resist that was **not** bombarded with ions. The wafer and most other layers on it will not be affected by this process. Molybdenum however will get oxidized, so is not suitable for plasma ashing with O₂.

Note that resist ashing **does NOT clean** the wafer! Visually the wafer may look clean, but on the edges of the structures resist residues will remain. These can only be observed with a SEM, and must be removed by wet cleaning before continuing with the next process step.

22. CLEANING: HNO₃ 99% and 69.5 %

Both sides

Clean 10 minutes in fuming nitric acid at ambient temperature. This will dissolve organic materials.
Use wet bench "HNO₃ 99% (Si)" and the carrier with the white dot.

Rinse Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.

Clean 10 minutes in concentrated nitric acid at 110 °C. This will dissolve metal particles.
Use wet bench "HNO₃ 69,5% 110C (Si)" and the carrier with the white dot.

Rinse Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.

Dry Use the "Avenger Ultra-Pure 6" rinser/dryer with the standard program, and the white carrier with a red dot.

23. WET ETCHING: Native oxide removal with Marangoni drying

Front side

Etch Read the Marangoni manual and follow all the instructions.
Use the Marangoni wet etch & drying system for **Si wafers**.
After rinsing the bath with DI water, fill it up with a "0.55% HF" solution (at room temperature), and lift up the wafer carrier.

Time Place the wafers in the carrier slots and lower them into the solution for 4 minutes.

Rinse Open the DI water valve to rinse the wafers for 5 minutes.

Dry One minute before lifting up the wafers, open the IPA valve.
Lift up the wafers and remove the dried wafers from the carrier slots.
Close all valves and lower the empty wafer carrier into the DI water.

Note: Metallization must be performed **within 72 hours**.

Alternative PIT step:

If the Marangoni system is out of order or not usable (not enough 0.55% HF solution available), then the native oxide can also be removed in the following way:

Etch	Use wet bench "0.55% HF" at ambient temperature, and the carrier with the black dot.
Time	4 minutes.
Rinse	Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.
Dry	Use the "Avenger Ultra-Pure 6" rinser/dryer with the standard program, and the white carrier with a black dot.

Note: Metallization must be performed **immediately** after drying (to prevent new SiO₂ formation).

25. METALLIZATION: 675 nm Al (with 1%Si)

Front side

Use the TRIKON SIGMA 204 sputter coater for the deposition of an aluminium metal layer on the wafers. The target must exist of 99% Al and 1% Si, and deposition must be done at 350 °C with an Ar flow of 100 sccm. Follow the operating instructions from the manual when using this machine.

Use recipe **AlSi 675nm @ 350C** to obtain a 675 nm thick layer.

Visual inspection: the metal layer must look shiny.

PIT explanation:

The desired aluminium layer thickness is actually 600 nm. But because later in the process a **FENCE REMOVAL** step must be performed where 75 nm of the layer will be lost, 675 nm is sputtered to compensate for this loss.

During thermal processes (e.g. alloying) 1% of the silicon from the wafer will diffuse into the aluminium layer. This causes voids in the Si that are filled up by the aluminium. This is called **SPIKING**, and results in bad or non working devices. To prevent spiking, Al(1%Si) is used for sputtering instead of pure Al. As the deposited aluminium layer is now already saturated with 1% silicon, no more silicon from the wafer can dissolve into the aluminium.

26. COATING

Front side

Use the coater station of the EVG120 system to coat the wafers with photoresist. The process consists of:

- A treatment with HMDS (hexamethyldisilazane) vapor, with nitrogen as a carrier gas, to improve resist adhesion.
- Spin coating of Shipley SPR3012 positive resist, dispensed by a pump.
- A soft bake (a.k.a. pre-bake) at 95 °C for 90 seconds.
- An automatic Edge Bead Removal (EBR) with a solvent.

Always check the relative humidity (48 ± 2 %) in the room before coating, and follow the instructions for this equipment.

Default use program "**1 - Co - 3012 - 1.4**",
or use program "**1 - Co - 3012 - 1.4 - noEBR**" when the EBR option is out of order.

27. ALIGNMENT AND EXPOSURE

Front side

Processing will be performed on the ASML PAS5500/80 automatic wafer stepper. Follow the operating instructions from the manual when using this machine.

Expose **mask IC** with job **Diesize-10mm\g10a-1 (layer ID 1)**. Use the correct exposure energy (check the energy table).

28. DEVELOPING

Front side

Use the developer station of the EVG120 system to develop the wafers. The process consists of:

- A post-exposure bake at 115 °C for 90 seconds, to prevent **standing waves** on the sidewalls of the resist structures.
- Developing with Shipley MF322 with a single puddle process.
- A hard bake (a.k.a. post-bake) at 100 °C for 90 seconds.

Always follow the instructions for this equipment.

Use program "**1 - Dev - SP**".

29. INSPECTION

Front side

Visually inspect the wafers through a microscope:

- Check if the correct mask (from the correct mask set) is exposed.
- Check if there are no resist residues on the exposed areas; these are NOT allowed.
- Check the overlay of the exposed pattern (if the mask was aligned to a previous pattern on the wafer).
- Check the linewidth of the structures.

30. PLASMA ETCHING: 675 nm Aluminium (sputtered at 350 °C)

Front side

Use the Trikon Omega 201 plasma etcher.

Follow the operating instructions from the manual when using this machine.

It is **not** allowed to change the process conditions and times from the etch recipe!

Use sequence **AL06_350** (with a platen temperature of 25 °C) to etch the aluminium layer.

Process conditions from chamber recipe AL06_350: (use for 0.6 - 1.0 µm Al sputtered at 350 °C)						
Step	Gasses & flows	Pressure	Platen RF	ICP RF	Platen temp.	Etch time
1. breakthrough	HBr/Cl ₂ = 40/30 sccm	5 mTorr	50 W	500 W	25 °C	endpoint
2. bulk etch	HBr/Cl ₂ = 40/30 sccm	5 mTorr	40 W	500 W	25 °C	endpoint
3. overetch	HBr/Cl ₂ = 30/15 sccm	5 mTorr	40 W	500 W	25 °C	60% of bulk

Note: Always perform an "ALUMINIUM FENCE REMOVAL" step after dry etching of aluminium!

INSPECTION: No aluminium residues or undercut are allowed on the etched areas.

PIT explanation:

During plasma etch processes ion bombardment takes place. When aluminium is plasma etched, some Al atoms will be sputtered and bury themselves into the sidewall of the resist layer. When ashing the resist after the etch process, the Al containing sidewalls can not be removed by the O₂ plasma. This causes residues at the edge of the structures, that look like a fence. These must be removed before proceeding with the next process step.

During plasma etching of Al (1% Si) the 1% silicon in the layer will be simultaneously etched with the Al, due to the poor selectivity of the process. The gasses that are used for etching the Al are also capable of etching Si.

During wet etching of Al(1% Si) the 1% silicon will **not** be etched. It will deposit on the wafer surface and is visible as a dull brown coloured layer. This causes a risk for short circuit, and therefore the Si must be removed in a separate wet poly-Si etch step. After wet etching of Al(1% Si) **no FENCE REMOVAL** step is needed.

31. ALUMINIUM FENCE REMOVAL

Front side

Plasma strip Use the Tepla Plasma Gigabatch 360 system to remove the photoresist in an oxygen plasma. Follow the instructions specified for the Tepla stripper, and use the quartz carrier. Use **program 1**: 1000 watts power and automatic endpoint detection + 2 min. overetching.

Moisten Rinse for 1 minute in wet bench "H₂O/Triton X-100 tby Al. Ets". Use the carrier with the yellow dot. The bath contains 1 ml Triton X-100 per 5000 ml deionized water.

Etching Use wet bench "Al. ets 35°C", and the carrier with the yellow dot.
1 liter buffered aluminium etch fluid contains:
770 ml concentrated phosphorus acid (H₃PO₄ 85%), 19 ml concentrated nitric acid (HNO₃ 65%),
140 ml concentrated acetic acid (CH₃COOH 100%) and 71 ml deionized water.

Etch time 30 seconds. 75 nm of aluminium will be lost.

QDR Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.

Drying Use the "Avenger Ultra-Pure 6" rinser/dryer with the standard program, and the white carrier with a black dot.

Note: An "ALUMINIUM FENCE REMOVAL" step must always be performed after dry etching of aluminium!

32. CLEANING: HNO₃ 99% metal

Front side

- Clean 10 minutes in fuming nitric acid at ambient temperature. This will dissolve organic materials. Use wet bench "HNO₃ 99% (**metal**)" and the carrier with a red and white dot.
- Rinse Rinse in the Quick Dump Rinser with the standard program until the resistivity is 5 MΩ.
- Dry Use the "Avenger Ultra-Pure 6" rinser/dryer with the standard program, and the white carrier with a black dot.

Note **Do not perform a "HNO₃ 69,5% 110C (Si)" cleaning step, as that will remove the metal layer!**

33. ALLOYING

Front side

Furnace tube: **A3** Program name: **EKL-FGA-400-NOP** Total time: ±50 min

PROCESS	TEMPERATURE (in °C)	GASSES & FLOWS (in liter/min)	TIME (in minutes)	REMARKS
boat in	400	nitrogen: 3.0	5	
stabilize 1	400	nitrogen: 3.0	5	
stabilize2	400	nitrogen: 3.0	10	
alloying	400	nitrogen: 3.0 hydrogen: 0.3	20	
purge	400	nitrogen: 3.0	5	
boat out	400	nitrogen: 3.0	5	

Note **Before starting the alloying process, always check if the temperature of the furnace is at 400°C.**
If this is not the case, then first run the Standby_400 recipe to go that temperature. This may take a while. If the temperature is higher, then the Aluminium layer will melt.

PIT explanation:

Alloying is performed to **improve the contact resistance** between the metal layer and the (doped) silicon from the wafer by removing the interface traps (there is no covalent bond between the Al and Si atoms). An alloy is a mixture of two different metals. In this case a mixture of aluminium and silicon (a semi metal) at the layer interface.

The process uses Forming Gas and is a heat treatment process to improve material properties by annealing in a reducing atmosphere, typically a mixture of hydrogen and an inert gas like nitrogen. This process is particularly effective at reducing defects, passivating surfaces, and lowering contact resistance.

Forming gas is a mixture primarily composed of hydrogen (H₂) and nitrogen (N₂). Typically, it's a blend of a small percentage of hydrogen (often 5%) and a larger percentage of nitrogen (usually 95%). The specific ratio can vary depending on the application.